So, this is one of those cases where cache doesn’t really help us a lot. As we mentioned in chapter 6, there aren’t many such situations, but still, sometimes that’s the case.

Reasoning in 2-D, we might try a different approach, inspired by real maps: divide our map in tiles, using a regular grid; this way, we can easily find which tile is a point from its coordinates (simply divide the value of each coordinate by the size of a tile, see figure 8.2) and search the closest points in the same tile, or in the neighboring tiles. This, indeed, seems to help reducing the number of points we need to compare; there is, however, a catch: this approach works if data is regularly spaced, which is usually not the case with real datasets, as shown in figure 8.2.

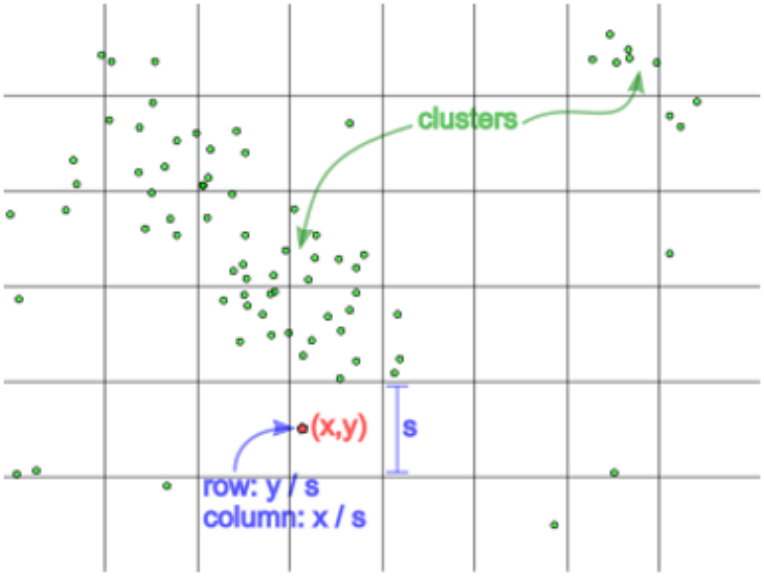


Figure 8.2 Indexing 2-D space with regular, equal-sized tiles. While finding the tile where a point lies is easy, irregular distribution of points in dataset causes, on average, many cells to be empty, while a few of them have high density.